What Drives Conversion in Quick Commerce?

An End-to-End Product Analytics Investigation

Harsha Narasimhe Gowda · linkedin.com/in/harsha-ng · April 2026

Tools: Google BigQuery · SQL · Python · Domain: Quick Commerce · Product Analytics · Experimentation

Executive Summary

User activity shows a clear **weekday vs weekend pattern**, with 30–40% lower engagement on weekends. The primary funnel bottleneck is the **Active → Cart stage**, but conversion improved by **+14 percentage points** from August to September — an improvement that warrants investigation. Checkout performance is strong at **65–73% cart-to-order**. High-intent placements drive conversion while discovery placements drive volume, revealing a clear strategic trade-off.

Why I Built This

Quick commerce is one of the most analytically interesting product categories in consumer tech. Users make decisions in seconds. Sessions are short, intent is high, and the margin between a great experience and a lost order is razor-thin.

I obtained a sample of anonymised quick commerce event data and used it to investigate a question I kept coming back to: **what actually drives conversion in a quick delivery app, and why does it change?**

This project follows a central mystery: a **14 percentage point improvement in funnel conversion** that appeared mid-period with no obvious explanation. Every section of this analysis builds toward understanding it.

The Dataset

Three anonymised event tables covering a three-week period (Aug 24 – Sep 14, 2025):

Table	Description
daily_events	All user-generated app events — sessions, browsing, interactions
cart_events	Product add-to-cart events with placement metadata
order_events	Order placement events with nested product-level JSON

Scale: ~380,000 inter-event gaps analysed · 15 product placement surfaces · 22 days of behaviour

Data Preparation & Validation

Duplicate Handling

Before any analysis I ran a full validation pass and found:

- **113 duplicate event IDs** in the daily events table
- **1 duplicate event ID** in the cart events table

Root cause: the same events were ingested twice, consistent with a pipeline re-processing artifact. Resolution: retained the earliest occurrence of each duplicate, ensuring correct event sequencing and no double-counting downstream.

Structural Transformation

The order events table stored product details as nested JSON. I flattened it to one row per product per order, retaining order_id, anonymous_id, and event_timestamp — enabling consistent user-product level joins across all three tables.

Data Completeness Note

Order data for Sep 12–14 shows zero orders despite significant cart activity. This is almost certainly a pipeline completeness issue, not real user behaviour. These dates are flagged throughout the analysis and excluded from conversion calculations where relevant.

Part 1 — Understanding the Audience

Active Users per Day

Active user defined as any anonymous_id generating at least one event on a given day.

Date	Active Users	Date	Active Users
Aug 24	260	Sep 4	287
Aug 25	403	Sep 5	324
Aug 26	393	Sep 6	326
Aug 27	366	Sep 7 (Sun)	187 ▼
Aug 28	397	Sep 8	334
Aug 29 ★	415 (peak)	Sep 9	319
Aug 30	371	Sep 10	284
Aug 31 (Sun)	274 ▼	Sep 11	312
Sep 1	364	Sep 12	303
Sep 2	295	Sep 13	333
Sep 3	284	Sep 14 (Sun)	176 ▼

Key findings:

- Peak activity: 415 users on Aug 29 (Friday)
- Every weekend showed a consistent 30–40% drop in activity
- Activity stabilised between 280–330 users on weekdays after Sep 1

Weekend users are a different audience — lower volume, lower engagement. A weekend homepage optimised for discovery rather than efficiency could recover some of this gap.

Part 2 — The Central Mystery

The Funnel and the 14-Point Lift

Daily three-step funnel: Active → Cart → Order. Each step uses same-day joins — a user counts at a step only if they performed that action on the same calendar day.

Date	Active	Cart	Order	Active→Cart	Cart→Order	Overall
Aug 24	260	81	46	31.2%	56.8%	17.7%
Aug 25	403	187	130	46.4%	69.5%	32.3%
Aug 26	393	172	127	43.8%	73.8%	32.3%
Aug 27	366	190	138	51.9%	72.6%	37.7%
Aug 28	397	198	126	49.9%	63.6%	31.7%
Aug 29	415	200	146	48.2%	73.0%	35.2%
Aug 30	371	192	135	51.8%	70.3%	36.4%
Aug 31	274	99	58	36.1%	58.6%	21.2%
Sep 1 ▲	364	213	151	58.5%	70.9%	41.5%
Sep 2	295	177	120	60.0%	67.8%	40.7%
Sep 3	284	182	107	64.1%	58.8%	37.7%
Sep 4	287	173	125	60.3%	72.3%	43.6%
Sep 5	324	200	146	61.7%	73.0%	45.1%
Sep 6	326	206	148	63.2%	71.8%	45.4%
Sep 7	187	87	57	46.5%	65.5%	30.5%
Sep 8	334	192	128	57.5%	66.7%	38.3%
Sep 9	319	197	123	61.8%	62.4%	38.6%
Sep 10	284	180	128	63.4%	71.1%	45.1%
Sep 11	312	185	115	59.3%	62.2%	36.9%

Sep 12-14 excluded — order data incomplete (0 orders despite cart activity)

The Investigation

Cart-to-order conversion was stable throughout — 65 to 73% across the entire period. The checkout experience was not the problem. The bottleneck was always Active → Cart. Then, starting September 1, it improved dramatically.

Period	Active → Cart	Cart → Order	Overall
Aug 24 – Aug 31	~31–52%	~57–74%	~18–37%
Sep 1 – Sep 11	~57–64%	~59–73%	~38–45%
Lift	**+14pp**	**Stable**	**+~10pp**

Why did it improve? Ruling out the obvious:

- **Not a weekend effect.** The improvement holds across all weekdays in September, not just high-traffic days.
- **Not a volume effect.** September weekday traffic is slightly lower than August peaks, yet conversion is higher.
- **Not random noise.** The lift is consistent across 11 consecutive weekdays.

Two hypotheses remain:

1. A product or UX change was shipped around Sep 1 that improved homepage relevance or search quality
2. The user cohort shifted — September brought a higher share of returning users who already knew what they wanted

To distinguish between these: correlate the lift with deploy logs from around Sep 1 and segment new vs. returning users across both periods. If a feature change is identified, design a holdback experiment to quantify its causal contribution.

Proposed A/B Test

Hypothesis: a personalised homepage showing products based on past purchase history increases active-to-cart rate vs. generic category layout.

Parameter	Specification
Unit of randomisation	User (anonymous_id) — not session, to avoid within-user contamination
Primary metric	Active-to-cart rate
Guardrail metrics	Session duration, overall order conversion, revenue per session
Duration	Minimum 2 weeks to capture weekday/weekend variation
Key segments	New vs. returning users (returning users benefit more from personalisation)

Part 3 — Where Users Convert (and Why)

Product Placement Performance

Conversion calculated at user-product level using last-touch attribution — the cart event closest in time before the order receives purchase credit. Cart events after the order timestamp are excluded.

Placement	Cart Adds	Purchased	Conversion	Vol. Rank
Out-of-stock substitutes	**1,523**	**1,182**	**77.6%**	**8**
Last bought	**6,772**	**5,150**	**76.0%**	**5**
Search	23,770	15,953	67.1%	2
Cart recommendations	2,033	1,364	67.1%	7
Personalised recommendations	8,173	4,889	59.8%	3
Category browse	30,257	18,005	59.5%	1
Product detail page	4,991	2,786	55.8%	6
Deals	6,781	3,198	47.2%	4
Other surfaces*	~2,700	~950	~50%	—

** Surfaces with <200 cart adds — statistically unreliable, not recommended for optimisation decisions*

The Intent vs. Discovery Trade-off

Surface Type	Conversion	Volume	Strategic Implication
High-intent (substitutes, last bought)	76–78%	Low	Users know what they want. Little room to improve further.
Mid-intent (search, cart recs)	67%	High	Strong volume + conversion. Focus on relevance quality.
Discovery (category, recommendations)	59–60%	Highest	Largest revenue opportunity. Improving relevance = biggest impact.
Browsing (deals, swimlanes)	44–48%	Medium	Low intent. Content targeting needs work.

Highest ROI opportunity: Personalised recommendations (59.8% conversion, 8K cart adds). Meaningful volume with meaningful room to improve. A purchase-history-weighted algorithm could push this to 65%+, generating hundreds of additional purchases per period.

Part 4 — Which Products Drive Conversion?

Conversion calculated at user-product level. Minimum threshold: 10 distinct users per product to filter noise. Purchase must occur after the add-to-cart event.

Product SKU	Users Added	Users Purchased	Conversion
11015448	15	15	100.0%
117801750	14	14	100.0%
13134784	13	13	100.0%
113651651	11	11	100.0%
11015675	21	20	95.2%
11018157	17	16	94.1%

The 100% Conversion Problem

Several products show 100% conversion. The catch: sample sizes are 10–21 users. These are almost certainly staple items — things users add to cart with full intent (milk, eggs, a specific brand). Important to stock reliably, but not the growth story.

A Better Definition of Best

Conversion rate x cart add volume = expected purchases per period. A product with 70% conversion and 1,000 cart adds generates 700 expected purchases. A product with 100% conversion and 10 cart adds generates 10.

The real growth opportunity is mid-conversion, high-volume products — items frequently added but not always purchased. Understanding why (price, stock, delivery fee friction) is where the product work lives.

Part 5 — Defining a Session

Sessions are not defined in the raw data — they need to be constructed from inter-event gaps. I analysed the gap distribution between consecutive events per user:

Gap Range	Event Count	% of Total
Under 1 minute	329,543	86.4%
1–5 minutes	27,387	7.2%
5–15 minutes	4,549	1.2%
15–30 minutes	2,690	0.7%
30–60 minutes	2,231	0.6%
Over 60 minutes	15,057	3.9%

86.4% of gaps are under one minute. User activity happens in short, intense bursts. After 30 minutes of inactivity, the behavioural signal is clear: the session is over.

Session definition:

A continuous sequence of user events where no consecutive gap exceeds 30 minutes. A session also ends upon order placement, since that represents the completion of user intent.

What Sessions Unlock

- Session-to-order rate — a more precise conversion metric than daily active users
- Events per session — engagement proxy separating high and low intent visits
- Entry point analysis — do search-entry users convert differently to homepage-entry users?
- Re-engagement patterns — how often do users browse in one session and order later?

Summary of Recommendations

Priority	Action	Expected Impact
P1	Investigate Sep 1 conversion lift — identify the driver and run a holdback test	Confirm and replicate the +14pp funnel improvement
P2	A/B test personalised recommendation algorithm (purchase-history weighted)	Estimated 5–8pp lift in recommendation-to-cart rate
P3	Diagnose mid-conversion, high-volume product drop-off	Identify stock, pricing, or friction issues hiding in the data
P4	Weekend-specific homepage strategy	Recover 30–40% weekend engagement gap
P5	Implement session-level analytics as primary reporting framework	Enable session-to-order rate as the core conversion KPI

This project was built independently using anonymised event data as a self-directed investigation into quick commerce product analytics. All SQL queries are available in the accompanying queries.sql file.

Harsha Narasimhe Gowda · harshangcr7@gmail.com · linkedin.com/in/harsha-ng · April 2026